

CHATTANOOGA, TN, USA

WMO: 723240

Lat: 35.034N

Lon: 85.200W

Elev: 204

StdP: 98.90

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13882

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-4.8	-15.5	1.0	-3.3	-13.2	1.2	-1.4	9.2	5.5	8.4	5.2	3.3	330	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	34.9	23.3	33.7	23.2	32.5	23.0	25.3	31.3	24.8	30.5	24.3	29.7	3.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.0	27.4	23.2	18.4	26.9	22.7	17.9	26.6	78.1	31.4	75.9	30.6	73.9	29.9	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.1	DB	-10.4	36.2	2.9	1.9	-12.5	37.6	-14.2	38.7	-15.8	39.8	-17.9	41.1	(4)
(5)			WB	-11.5	26.2	2.7	0.6	-13.4	26.7	-15.0	27.1	-16.5	27.4	-18.5	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	5.2	7.4	11.7	16.6	21.0	25.1	26.7	26.5	23.2	17.0	10.7	6.9	(6)	
	DBStd	8.67	5.56	5.15	4.90	4.15	3.49	2.44	1.90	2.10	3.15	4.25	4.46	4.94	(7)	
	HDD10.0	483	169	101	39	3	0	0	0	0	0	3	44	123	(8)	
	HDD18.3	1687	409	306	211	81	17	0	0	0	4	75	231	354	(9)	
	CDD10.0	2877	19	28	92	201	342	454	518	513	397	221	66	27	(10)	
	CDD18.3	1040	0	1	6	28	100	204	259	254	150	34	2	1	(11)	
	CDH23.3	9646	0	3	54	292	893	1898	2532	2443	1239	274	16	1	(12)	
(13)	CDH26.7	3918	0	0	6	58	284	806	1121	1097	485	58	1	0	(13)	
(14)	Wind	WSAvg	2.2	2.6	2.7	2.9	2.6	2.2	1.9	1.8	1.7	1.7	1.9	2.0	2.3	(14)
(15) Precipitation	PrecAvg	1362	125	118	143	115	103	98	126	90	108	84	119	132	(15)	
	PrecMax	1871	249	247	414	303	284	233	303	192	360	182	243	348	(16)	
	PrecMin	891	23	18	30	18	28	16	20	0	13	2	39	22	(17)	
	PrecStd	253	53	59	81	64	54	50	68	41	79	52	49	67	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	23.5	27.5	30.3	32.6	35.4	36.2	36.8	35.5	30.6	25.6	22.2	(19)	
		MCWB	16.1	16.7	17.4	19.2	21.7	23.1	23.9	23.8	21.4	20.6	17.7	17.0	(20)	
	2%	DB	18.5	20.7	24.8	28.4	31.1	33.8	34.4	34.9	33.3	28.2	22.7	19.5	(21)	
		MCWB	14.6	14.9	16.3	18.2	21.0	23.1	24.0	23.6	21.6	19.6	15.8	16.0	(22)	
	5%	DB	16.4	18.5	22.6	26.7	29.8	32.5	33.3	33.4	31.4	26.4	20.6	17.1	(23)	
		MCWB	13.0	12.8	14.6	17.4	20.6	22.7	23.8	23.3	21.6	18.6	15.3	13.9	(24)	
(25) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	14.0	16.1	20.3	24.8	28.2	31.1	32.0	32.0	29.6	24.5	18.5	14.9	(25)	
		MCWB	11.0	11.4	13.6	16.7	20.2	22.3	23.6	23.1	21.3	18.0	14.1	12.1	(26)	
	0.4%	WB	17.5	18.1	19.1	21.5	23.6	25.2	26.2	26.0	24.1	22.9	19.9	18.9	(27)	
		MCDB	19.4	21.6	24.3	27.0	29.2	31.6	32.0	32.4	29.4	27.6	22.1	20.9	(28)	
	2%	WB	15.8	16.5	17.7	20.2	22.7	24.6	25.4	25.3	23.5	21.6	18.1	16.6	(29)	
		MCDB	18.2	19.8	22.6	25.5	28.4	30.6	31.2	31.5	28.6	25.9	20.8	18.7	(30)	
(31) Mean Daily Temperature Range	5%	WB	13.8	14.5	16.4	19.0	21.9	24.0	24.9	24.7	23.0	20.5	16.7	14.7	(31)	
		MCDB	15.9	17.0	20.6	24.1	27.4	29.7	30.5	30.4	27.9	24.1	19.6	16.5	(32)	
	10%	WB	11.5	12.4	15.0	18.0	21.2	23.4	24.4	24.1	22.4	19.4	14.8	12.5	(33)	
		MCDB	13.3	15.1	19.0	22.7	26.1	28.8	29.8	29.3	26.9	23.0	17.7	14.5	(34)	
	5% DB	MDBR	9.9	10.5	11.6	12.5	11.6	10.7	10.1	10.4	10.9	11.8	11.5	9.6	(35)	
		MCDBR	11.9	13.9	15.0	14.8	13.4	12.5	11.7	12.3	13.4	13.6	14.3	11.7	(36)	
(37) Clear-Sky Solar Irradiance	5% WB	MCWBR	8.3	8.5	7.4	6.3	4.8	4.0	3.4	3.5	4.1	5.9	8.2	8.0	(37)	
		MCWBR	9.9	11.0	11.9	11.5	11.0	10.5	10.2	10.3	9.8	9.8	10.3	9.6	(38)	
	taub	taub	0.322	0.334	0.364	0.411	0.465	0.504	0.549	0.550	0.438	0.380	0.343	0.318	(40)	
		taud	2.486	2.448	2.395	2.250	2.136	2.065	1.960	1.939	2.241	2.395	2.442	2.518	(41)	
(42) All-Sky Solar Radiation	Ebn at Noon	882	910	907	875	829	794	755	744	820	845	846	865	865	(42)	
	Edh at Noon	85	98	112	136	155	166	183	184	130	102	88	78	78	(43)	
(44) All-Sky Solar Radiation	RadAvg	2.40	2.94	4.08	5.27	5.82	6.17	5.84	5.45	4.68	3.81	2.76	2.05	2.05	(44)	
	RadStd	0.18	0.38	0.38	0.35	0.44	0.45	0.37	0.29	0.43	0.49	0.32	0.20	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+168	+120
(47)	Regional (34 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+103

Nomenclature: See separate page